Ayaan Monappa

amonappa7@gatech.edu | 404-901-5884 | Linkedin: Ayaan Monappa | Portfolio

Education

Georgia Institute of Technology **Atlanta, GA**

Bachelor of Science in Computer Engineering GPA: 3.87 *Expected Graduation: May 2028*

Courses: ECE 3550 (Feedback Control Systems), ECE 2026 (Signal Processing), ECE 2035 (Assembly and C), CS 1332 (Data Structures & Algorithms), ECE 2040 (Circuit Analysis), ECE 2031 (Digital Design)

Clubs: Bioinformatics Researcher@GT, Medical Robotics@GT, GT Club Baseball, Georgia Tech International Society.

Dean's List **2025, 2026**

Experience

Solutions Engineer Intern - Avetics Drone Technologies **May 2026 - Aug 2026**

- Built an autonomy stack and engineered the unmanned boat's power and telemetry circuitry into a working prototype capable of waypoint navigation; iterated feature development over open-water trials.
- Developed and integrated a LiDAR-based obstacle avoidance system, designed to let the vessel to navigate around obstacles autonomously by implementing point cloud transformation and filtering with 4 ROS 2 nodes
- Engineered an encrypted communications network: **a 500 m** Wi-Fi mesh and 4G connectivity, enabling low-latency video transmission, telemetry monitoring, and remote operation.
- Trained a PyTorch machine learning model capable of identifying 9 classes, reaching **95%+ mean average precision** onboard the prototype to enable real-time detection in the Singapore Strait.

Undergraduate Researcher - AutoRobotics in Construction Lab **Aug 2025 - May 2026**

- Processed ~**25,000 samples** of raw wristband biosignals from exoskeleton-wearing construction workers, filtering noise, normalizing per participant and categorizing data to build the set for model training and validation.
- Trained and validated **4 classifiers** to predict worker frustration and mental demand, achieving **91% accuracy** and **0.86 macro-F1** on a model that powers a live risk dashboard for site supervisors.

Electrical Subteam Member - Georgia Tech Medical Robotics **Sep 2025 - May 2026**

- Designed Arduino circuit and firmware for a posture wearable, including a flyback-protected motor driver, filtered roll/pitch algorithms and a CAD enclosure; demoed to **100**+ industry professionals.
- Built the power budget and protection circuit for a transradial upper-limb prosthetic.

Software Engineer - Cravr Salus Inc **Dec 2025 - Feb 2026**

- Shipped Swift games to production in a behavioral-health iOS app with **50,000**+ users, helping them interrupt stress binge-eating urges.
- Added new databases to the existing Firebase backend to serve dynamic game content.

Projects

Cofounder - Modular Robotics **Sep 2026 - Present**

- Designing an autonomous modular bot that splits into smaller agents to work on a task in parallel, then recombines to travel; defining the docking mechanism, power, and autonomy stack.

Grasp - HackATL 2026 **Oct 2026**

- Co-designed and pitched Grasp, a gesture- and voice-controlled 3D visualization tool using laptop-webcam hand tracking; defined the software stack and simulated the product, competing among **400** participants.

Founder - Glowin Tech **Mar 2024 - Feb 2026**

- Built a full-stack iOS fitness app and launched it to public beta, writing ~**5,500 lines** of Swift across a SwiftUI client and a Vapor server that handled **6,000**+ API calls in beta.
- Designed a REST API with ~**21** endpoints and built a computer vision push-up form checker.

Skills

Languages: Python, C, Java, Swift, SQL, Bash, RISC-V Assembly
Robotics: ROS 2, ArduPilot, BlueOS, MAVLink/MAVROS, LiDAR, GStreamer
Libraries: PyTorch, scikit-learn, OpenCV, NumPy, pandas
Embedded & Hardware: Altium, Arduino, ESP32, KiCad, LTspice, Fusion 360, Linux, oscilloscope
Manufacturing: Soldering, crimping, 3D printing, aluminum-extrusion framing, thread tapping
Online Coursework: Visual Navigation for Autonomous Vehicles (MIT), Battery Management Systems (TU Delft)